

MILES CITY, MT, USA

WMO: 742300

Lat: 46.427N

Lon: 105.883W

Elev: 800

StdP: 92.08

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24037

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.3	-22.8	-29.5	0.3	-25.9	-26.0	0.4	-22.6	14.0	-0.3	12.0	-2.9	3.1	150	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	36.7	18.5	34.7	18.1	32.8	17.8	20.8	31.2	20.0	30.3	19.1	29.2	4.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	14.0	24.0	16.7	13.1	23.5	15.6	12.2	22.9	63.9	31.0	60.5	30.3	57.3	28.8	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.6	9.0	-30.8	40.1	4.0	1.8	-33.7	41.4	-36.0	42.5	-38.2	43.5	-41.1	44.8		
(5)			-30.7	22.8	3.5	1.3	-33.2	23.7	-35.3	24.5	-37.3	25.3	-39.9	26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	-6.5	-4.6	1.4	7.8	13.3	18.9	24.2	23.0	16.7	8.5	0.7	-5.3		
	DBStd	12.20	7.90	8.20	7.52	5.26	4.72	4.18	3.56	3.82	5.13	5.23	6.82	7.44		
	HDD10.0	2167	512	409	272	99	22	1	0	0	7	89	281	473		
	HDD18.3	4128	771	642	524	316	167	42	2	8	89	307	528	732		
	CDD10.0	1526	0	0	6	34	122	268	439	402	209	42	3	0		
	CDD18.3	447	0	0	0	1	9	59	183	152	41	2	0	0		
	CDH23.3	5923	0	0	3	33	176	751	2361	1937	614	48	1	0		
	CDH26.7	2871	0	0	0	6	43	293	1251	998	266	13	0	0		
(14) Wind	WSAvg	4.3	4.1	4.2	4.5	4.8	4.8	4.5	4.1	4.1	4.2	4.4	4.1	4.1		
(15) Precipitation	PrecAvg	344	12	10	16	36	55	70	42	26	30	24	12	13		
	PrecMax	515	45	30	45	107	173	164	160	99	102	160	55	45		
	PrecMin	134	1	0	0	0	7	18	0	0	0	0	0	0		
	PrecStd	98	10	8	10	27	35	38	37	22	26	26	11	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	15.7	22.5	27.2	30.6	36.1	39.5	38.6	35.7	28.2	19.8	12.3		
		MCWB	5.4	6.8	10.5	13.1	16.8	18.6	19.6	19.2	17.7	14.5	9.6	5.4		
	2%	DB	7.5	11.0	18.6	23.1	27.5	32.5	37.3	36.2	32.7	23.5	15.5	8.7		
		MCWB	3.0	4.7	8.9	11.7	15.1	18.5	18.9	18.1	16.5	12.2	7.6	3.6		
	5%	DB	5.2	7.7	14.8	20.1	25.1	30.0	35.2	34.3	29.9	20.2	12.5	6.0		
		MCWB	1.9	3.3	7.0	10.4	14.4	17.8	18.5	17.6	15.7	10.9	6.0	1.8		
	10%	DB	3.5	5.1	11.7	17.4	22.6	27.7	33.2	32.4	27.0	17.3	9.9	3.8		
		MCWB	0.8	1.8	5.6	8.9	13.2	16.8	18.3	17.4	14.6	9.6	4.7	0.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	7.6	11.0	14.1	18.6	21.6	22.5	21.3	19.6	14.9	10.1	5.8		
		MCDB	11.3	14.9	21.1	25.1	26.2	30.5	32.6	33.3	29.7	26.2	18.6	11.3		
	2%	WB	3.4	5.2	9.3	12.3	16.7	20.1	21.2	20.1	17.9	13.0	8.0	3.8		
		MCDB	7.0	10.2	17.6	21.7	24.6	28.9	31.8	31.2	28.1	21.5	14.9	8.3		
	5%	WB	2.0	3.5	7.5	10.8	15.2	18.9	20.3	19.2	16.7	11.5	6.4	2.1		
		MCDB	4.7	7.5	14.5	19.0	22.9	27.6	30.8	29.7	27.3	18.9	12.1	5.6		
	10%	WB	0.9	1.9	5.8	9.5	13.9	17.7	19.5	18.4	15.3	10.1	4.9	0.7		
		MCDB	3.3	4.8	11.2	16.5	21.2	26.1	30.0	28.8	25.4	16.7	9.5	3.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.6	11.0	12.2	12.9	12.8	13.4	15.2	15.4	14.8	12.9	11.9	10.6		
		MCDBR	11.6	13.7	17.8	17.7	16.8	17.0	17.8	18.3	19.3	18.1	15.9	12.8		
		MCWBR	8.1	8.6	9.6	8.7	7.6	6.5	5.8	6.2	7.5	8.9	9.2	8.5		
		MCDBR	11.2	12.9	17.0	16.2	14.9	15.0	15.6	15.6	16.7	16.4	15.5	12.5		
	5% WB	MCWBR	8.2	8.3	9.4	8.3	7.1	6.5	5.9	5.9	6.8	8.3	9.1	8.6		
(40) Clear-Sky Solar Irradiance	taub		0.251	0.258	0.273	0.308	0.332	0.345	0.357	0.360	0.320	0.285	0.259	0.247		
	taud		2.569	2.574	2.579	2.464	2.456	2.453	2.440	2.399	2.500	2.588	2.572	2.561		
	Ebn at Noon		883	944	969	951	933	919	902	885	899	895	863	849		
	Edh at Noon		59	72	83	103	108	109	109	109	90	71	58	53		
(44) All-Sky Solar Radiation	RadAvg		1.62	2.57	3.74	4.95	5.90	6.81	7.23	6.01	4.47	2.86	1.82	1.29		
	RadStd		0.14	0.18	0.32	0.26	0.57	0.45	0.26	0.33	0.32	0.22	0.18	0.09		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page